YEAR 10 MATHS · CHAPTER 2 · MIND MAP

Geometry and networks

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1 Review of geometry

Interior angle sum of an n-gon: $(n-2) \times 180°$. Exterior angles add to 360°.

Regular octagon: exterior $360° \div 8 = 45°$, interior $135°$

2 Congruent triangles and quadrilaterals

Tests: SSS, SAS, AAS, RHS. Use them to prove properties of shapes.

In a parallelogram, the diagonals cut it into congruent triangles (AAS), so the diagonals bisect each other

Watch out: AAA shows similar, not congruent.

3 Similar figures and triangles

Scale factor k: lengths $\times k$, areas $\times k^2$, volumes $\times k^3$.

$k = 3$: area $\times 9$, volume $\times 27$

Similar triangles, matching sides 4 and 10: $k = 2.5$, so side 6 matches 15

4 Circles and chords

A line from the centre at right angles to a chord bisects the chord.

Chord 16, radius 10: distance from the centre $\sqrt{10^2 - 8^2} = 6$

Chord, arc, sector, segment, tangent, secant: know each word.

5 Angle properties of circles

Angle at the centre is twice the angle at the circumference. Angles in the same segment are equal.

Angle in a semicircle = 90°

Cyclic quadrilateral: opposite angles add to 180° (110° opposite 70°)

Circumference angle 35°: centre angle 70°

6 Circles and tangents

A tangent meets the radius at 90°. Two tangents from one point are equal.

Radius 5, centre 13 away: tangent $\sqrt{13^2 - 5^2} = 12$

Alternate segment theorem: tangent–chord angle equals the angle in the opposite segment.

7 Chords, secants and tangents

Chords: $AP \times PB = CP \times PD$. Tangent: $PT^2 = PA \times PB$.

$4 \times 6 = 3 \times x \Rightarrow x = 8$

$PA = 4,\ PB = 9$: $PT = \sqrt{36} = 6$

8 Introduction to networks

Vertices (dots) joined by edges. The degree of a vertex is the number of edges at it.

The degrees always add to twice the number of edges

5 edges: degrees add to **10**

9 Isomorphic and planar graphs

Isomorphic: same connections, drawn differently. Planar: can be drawn with no crossing edges.

Euler's formula for planar graphs: $v - e + f = 2$

4 vertices, 6 edges: $f = 2 - 4 + 6 = 4$ faces

10 Trails, paths and Euler circuits

Trail: no repeated edges. Path: no repeated vertices.

Euler circuit: every vertex has an even degree

Euler trail: exactly two odd vertices (start at one, finish at the other)

11 Shortest path problems

Add the weights along each route and keep the smallest at every vertex.

A to B 4, B to D 3, A to C 2, C to D 6

A–B–D = 7, A–C–D = 8, so the shortest is **7**